

Project Title: Deriving insights for research data sharing from a large-scale data sharing initiative in Canada

1 Recruitment and consent

Will this study involve recruitment of human study participants?

☒ Yes

☐ No

How are potential study participants identified and/or recruited to the study?

Explain how potential participants are identified or introduced to the study, and who will recruit participants. Will the investigator/s require any special permissions or access to the target population e.g. clinic access, patient registries or records, mailing lists, community access?

This study investigates the organizational dimensions of health data sharing by analyzing the administrative correspondence generated during the creation and operation of the COVID-19 Immunity Task Force (CITF) Databank — a pan-Canadian initiative that assembled health records from over 100 COVID-19 studies across Canada. The CITF was an initiative whose mandate was to catalyze, support, fund and harmonize knowledge on SARS-CoV-2 immunity for federal, provincial, and territorial decision-makers in their efforts to protect Canadians and minimize the impact of the COVID-19 pandemic. The CITF Databank provides continued support to the research community by centralizing research data provided by CITF-funded studies and by making the data accessible for extended research. This study is funded through the CIHR Project Grant - Priority Announcement: Pandemic Preparedness and Health Emergencies Research (#587162), and was scientifically reviewed under the mainline Project Grant application (#582492).

The CITF Databank’s email archive is a dense record of the coordination work that made data sharing happen: who communicated with whom, about what, and when. Teams that contributed data comprised members with diverse professional roles and backgrounds, including researchers, trainees, administrative staff and technical support staff. The email archives and the multitude of related stakeholders who contributed to the CITF Databank afford great opportunities to understand the variety of practical needs and challenges that scaffold health research data-sharing initiatives. This study therefore draws from (1) emails sent between members of the core CITF Databank team and those who contributed data to the CITF Databank, and (2) a questionnaire directed toward the CITF Databank’s contributors.

The email corpus is derived from the archived mailboxes of key CITF Databank personnel, including the Scientific Director, Operations Manager and Data Managers, who are responsible for communicating with the data contributors. These individuals are data subjects of this study’s secondary-use analysis of the email corpus; they are not members of the research team, and TCPS2 CORE training does not apply to them. We have successfully tested the feasibility

of a workflow to process this data in a manner that complies with ethical guidelines, including de-identification processes, as outlined in the research protocol.

For the questionnaire, participants are identified through the CITF’s roster of data contributors, including project leaders, data managers and key technical and administrative personnel. They will be contacted using the email addresses that the CITF has maintained as part of its ongoing communication with its members. We may also ask project leaders to forward the questionnaire to additional team members who worked in a less visible capacity and whose contact information the CITF does not maintain.

The total sample size reported on this application reflects the combined scope of both study populations, not the number of partner studies. The CITF Databank centralizes data from over 100 partner studies, each typically represented by 2 to 3 contactable individuals (project leads, data managers, and technical or administrative personnel), yielding an estimated 200 to 400 individuals in the questionnaire’s target population. The email corpus separately captures correspondence involving a comparable or larger number of individuals, including senders and recipients beyond the questionnaire’s roster-based population, as described above. The reported sample size of 400 reflects this combined estimate.

Describe the consent process. *If alternate processes for seeking consent are planned (e.g. verbal, online, waiver), please provide a rationale and outline the procedure of obtaining and documenting consent and/or assent, where applicable.*

We seek a waiver of consent for the email data, on the grounds of secondary use of identifiable information under TCPS2 Article 5.5A. The email corpus was generated for the CITF Databank’s administrative operations, not for this research, and the research team members are not the original correspondents. We address each of the six criteria in Article 5.5A below.

1. Identifiable information is essential to the research. The study’s objective is to characterize the coordination processes and organizational determinants of the CITF Databank’s operations. This requires analysis of who communicated with whom, in what roles, and at what points in the data-sharing lifecycle, information that is only present in the identifiable correspondence itself. Aggregated or anonymized alternatives (e.g. summary statistics collected after the fact) would not capture the relational and temporal structure that is the object of study.
2. Proceeding without consent is unlikely to adversely affect participants’ welfare. All emails were sent in a professional, institutional capacity in the course of Databank operations, and do not constitute personal correspondence. Emails are de-identified prior to analysis, no email text is published, and analysis proceeds only on abstracted variables (topic, phase, gender, affiliation, role).
3. Appropriate measures will be taken to protect privacy and safeguard identifiable information. These are described in the Confidentiality and Data Security section, including de-identification procedures, McGill Cloud Directive-compliant storage, and restricted access to unprocessed records.

4. The researchers will comply with any known preferences previously expressed by individuals about use of their information. No mailbox holder or correspondent has expressed a preference restricting research use of their CITF Databank correspondence. Should any such preference come to light during the study, it will be honoured, including exclusion of the relevant correspondence from analysis.
5. Seeking consent is impossible or impracticable. The email corpus captures correspondence with a substantially larger and more heterogeneous population than the questionnaire's roster-based recruitment. Senders and recipients in the corpus include individuals no longer affiliated with their CITF-funded studies, personnel who interacted with the Databank in a peripheral or one-time capacity, and third parties such as institutional administrators, for whom the CITF Databank retains no current or reliable contact information. This population is not equivalent to the questionnaire's target population: the questionnaire is directed at current or known former project leaders, data managers, and technical/administrative personnel identified through the CITF roster, a substantially narrower and more contactable group. The two data sources are treated as analytically parallel and non-overlapping strands, compared only through abstracted categories (role, gender, affiliation), not through individual-level linkage; this separation is a deliberate privacy-protective design feature of the study, not an incidental gap. Where the research team is able to contact members of the roster for the questionnaire, this reflects the narrower and different population targeted by that separate work package, not evidence that consent from the full, more diffuse population of email correspondents is practicable.
6. Any other necessary permission for secondary use has been obtained. Access to the email corpus is authorized by the CITF Databank's Scientific Director, in his capacity as custodian of the Databank's operational records.

Individuals who are contacted for the questionnaire will be provided with an informed consent document that outlines in more detail the goals of the study, the roles of the participant, how they will be recorded, how data pertaining to them will be retained, and the potential risks and benefits pertaining to their involvement. This document will also describe how participants' personally identifiable information will be managed and used. This document will be attached to emailed solicitations to participate in the study, and will also be available as a preamble to the online questionnaire. Participants will be prompted to read the document and required to affirm their consent prior to beginning the questionnaire. Participants' names and affirmation of their consent will be retained only to document that they have consented to participate in the study, in case the project is audited.

Is there a relationship between the study participants and the person obtaining consent and/or the principal investigator/s?

- ☒ Yes
☐ No

If yes, please explain the nature of the relationship, and outline the steps that will be taken to avoid the perception of undue influence.

The project that serves as the case in this research is the Covid-19 Immunity Task Force (CITF) Databank. The principal investigator currently serves as the CITF Databank's Scientific Director. This study draws on two distinct sources: the email corpus, built from the archived mailboxes of three CITF Databank operational roles (the Scientific Director, the Operations Manager, and the Data Managers); and the questionnaire, whose participants are Databank contributors, i.e., individuals at CITF-funded partner studies who have received funding from the CITF and collaborate with the CITF Databank.

The PI holds no personal, financial, or professional interest that would constitute a conflict of interest. CITF funding for the partner studies represented among questionnaire participants has concluded, and the PI holds no ongoing funding or collaboration authority over them. The recruitment email and informed consent document both state explicitly that participation is voluntary and that a decision to decline or withdraw will have no bearing on the participant's standing with, or any past or future collaboration with, the CITF Databank.

Nonetheless, to avoid the perception of undue influence given the PI's institutional role, all participant-facing recruitment and consent activities are handled by Zachary Batist, Postdoctoral Researcher, who holds no authority over CITF funding decisions or collaboration status with the Databank.

2 Risk-benefit assessment

Describe the foreseeable risks to study participants. *What risks are attributable to the research, including cumulative risks? Which risks are participants normally exposed to in the course of their clinical care or in their daily activities as they relate to the research questions/objectives?*

Participation in this study does not involve any physical, psychological or legal risks.

The questionnaire may ask participants to share information about their work practices and work relationships. The only conceivable harm concerns potential disruption or complication of participants' professional reputations, however since no personal identifying information will be associated with their responses, the actual risk is extremely minimal. Likewise, all personal information is obfuscated in the emails data, which significantly minimized any risk of harm pertaining to that dataset as well.

In cases where participants provide detailed textual responses in the questionnaire, all personal information that they may include will be obfuscated prior to analysis. This will be made clear in the questionnaire response guidelines associated with those pertinent questions or prompts.

What procedures are in place to monitor and assess participant safety for the duration of the study?

Prior to responding to the questionnaire, participants will be required to affirm their consent to participate in the study. Participants will also be encouraged to reach out with any questions or concerns, and they will be provided contact information for Zachary Batist, the Postdoctoral Researcher who handles all participant-facing recruitment and consent activities for this study.

Although we strive to automate much of the emails data preprocessing, a human is always involved to verify and correct the integrity of these processes, especially the de-identification components.

Describe the potential benefits of the study for: *(1) the study participants; (2) the population under investigation, and (3) the field of research.*

This study contributes to the development of better epidemiological data-sharing infrastructures by articulating the administrative and operational aspects involved in such work. In particular, the study demonstrates how these factors relate to, overlap with or conflict with technical factors. By explicitly framing data sharing as more than a technical activity, the study informs the design of more effective data-sharing infrastructures that complement the institutional realities that contextualize effective health data sharing initiatives. This work therefore documents and reflects on the ways in which epidemiologists mobilize distributed records, and helps develop practical solutions that enable more effective collaborative workflows.

3 Privacy and confidentiality

Please describe the measures in place for meeting confidentiality obligations. *How is information and data safeguarded for the full cycle of the study: i.e. during its collection, use, dissemination, retention, and/or disposal?*

Even with the minimal risk profile associated with this study, it does not rely on personal identifying information at all.

All email records will be de-identified before any analysis or storage of results. De-identified emails are subjected to topic modelling to ascertain the main topics addressed in each. Non-personal metadata such as time stamps and broad-level institutional affiliations (derived from email address' domain names) are retained. Individuals' roles are inferred from their professional titles, but titles are not retained (roles characterize a person's responsibilities in relation to a project's collective objectives, and are not identifying information). Emails are therefore analyzed as relationships between a few abstracted variables (topics, phases, gender, affiliations, and roles) rather than records tied to any specific individual. As such, no identifying information will be included in the project's analysis.

The questionnaire will ask participants' about their gender, affiliation, and role, and questions or blocks of questions will be designed to correspond with topic and phase distinctions ascertained

in prior email analysis. There is no need to draw direct relationships between individual email senders and questionnaire respondents; the analysis focuses on aggregate experiences corresponding with roles, genders, and institutional hosts, and parity in our controlled vocabularies for the analysis of emails and questionnaire data allows for comparison of these distinct perspectival and documentary data sources.

The study is committed to adhering to fundamental data security practices, including those specified in McGill University’s Cloud Directive, which regulates the curation of sensitive research data. Physical records will be kept in a locked drawer in secure workspaces, either at McGill University’s School of Population and Global Health or at the principal researcher’s home office. Digital records will be stored on encrypted and password-protected drives and on secure servers approved or managed by McGill University under the Cloud Directive.

If a contracted cloud/storage service provider or online survey tool is used, provide information on the service provider’s security and privacy policy, location of its servers, data ownership, and what happens to the stored data after the contract is terminated. For more information, please consult the University’s directive.

The study uses file-sharing software hosted by the CITF Databank at McGill University’s School of Population and Global Health to backup all files maintained for this study. These backups include unobfuscated versions of email records containing potentially sensitive information, which exist in an unprocessed state. The software used to manage these backups is managed by McGill University and has been approved for storing sensitive research data by the Cloud Directive.

Access to these unobfuscated backups is restricted to Zachary Batist (Postdoctoral Researcher), David Buckeridge (Principal Investigator), and Research Assistants involved in the email processing pipeline. No other CITF Databank personnel have access. The backups are retained until project completion, at which point they are permanently deleted; only de-identified data is retained thereafter.

The study uses the secure GitLab instance hosted by the McGill Clinical and Health Informatics Research Group to store and track changes to research data and code. This software is managed by McGill University and has been approved for storing sensitive research data by the Cloud Directive.

The study will use LimeSurvey for distributing the questionnaire and managing the data it collects. LimeSurvey is approved by McGill’s Cloud Directive and we will use an instance hosted and managed by the university’s information technology division.

The study maintains a website where the team shares documentation and protocols that support the study. This is hosted using GitHub Pages and is backed up using Dropbox. No sensitive research data passes through these services.

Please explain any reasonable and foreseeable disclosure requirements (e.g. disclosure to third parties such as government agencies or departments, community partners in research,

personnel from an agency that monitors research, research sponsor, the REB/IRB, or regulatory agencies).

No disclosure requirements are foreseen.

If there are plans for retaining participant and/or study data for future use, please describe the context for its use, requirements for potentially re-contacting study participants and consent, and how the data will be stored and maintained for the long term.

Data will be published in compliance with ethical standards for sharing open research data. No identifying information will be retained. No identifying information will be included in any published data.

The principal investigator may invite select participants to collaborate on papers presenting the findings or advocating for actions based on the project's findings.

Secondary use of data studies: *if the study involves data linkage, please describe the data that will be linked and the likelihood that identifiable information will be created through the linkage.*

This project does not rely on data deriving from other studies. The data may be reused in related work being undertaken under the same grant and by those who access the openly accessible data after they are published.

4 Managing conflicts of interest

Conflicts of interest do not imply wrong-doing. It is the responsibility of the investigator to determine if any conflicts apply to any person/s involved in the design and/or conduct of the research study or any member of their immediate family. Disclose all contracts and any conflicts of interest (real, perceived, or potential) relating to this research project. Conflict of interest may also arise with regard to the disclosure of personal health information.

- ☒ Not applicable. There are no conflicts of interest to disclose.
- ☐ Yes, there are conflicts of interest to disclose.

If yes, please describe the conflicts of interest (real, potential, and perceived), and the procedures for managing declared conflicts. Not applicable. The PI holds no personal, financial, or professional interest that constitutes a conflict of interest. CITF funding for the partner studies represented among questionnaire participants has concluded, and the PI holds no ongoing funding or collaboration authority over them. Recruitment and consent for this study are nonetheless conducted independently by Zachary Batist, Postdoctoral Researcher, rather than by the PI.